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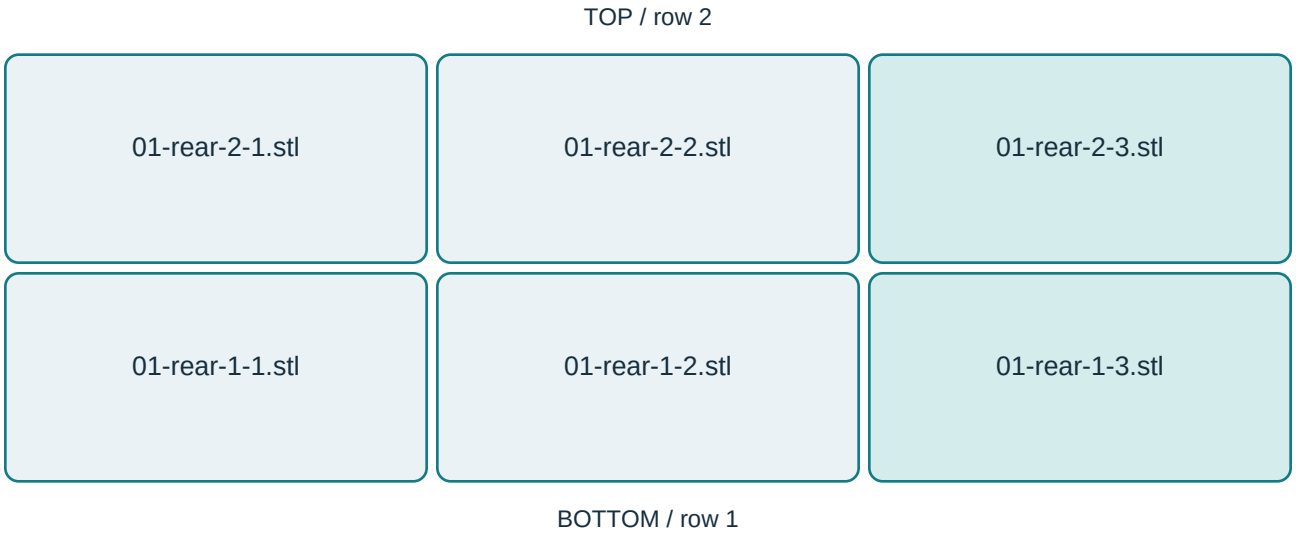
The full covered wall enclosure, with an independent right-hand Pi bay

This guide matches REV8: the Pi sits **beside the monitor in the 4.5-inch right bay**. The monitor and wiring are covered; the right lid opens separately for service. The latest corner-anchor print revision changes removable bed adhesion features only, not assembly.

1. Identify the six rear tiles

View from the FRONT into the empty chassis, with the wall behind it. Row 1 is the bottom row; row 2 is the top row. Column 3 forms the right-hand service bay. Rotate each printed tile into its assembled orientation before joining.



Layout diagram is schematic, not a drilling or scale template. Overall shell: 488.3 W x 269.5 H x 70 D mm, plus 8 mm wall spacers.

2. Sort the remaining parts

Group	Required printed pieces
Chassis and monitor	7 chassis splices; 2 bottom shoes; 2 rear posts; 1 left and 1 right side stop; selected rear shims.
Front and electronics	6 structural trim pieces; 1 open Pi carrier; 2 right cover halves; 2 cover splices; 2 cover hooks.
Wall and cable exits	4 wall spacers; 1 anti-lift bracket; 2 blank inserts; 1 cable insert; channel/cap pairs as needed.

Remove brims and corner ears after cooling. They are print aids, not assembly parts. Do not mix this kit with the old jigsaw basket or REV6/REV7 mounting arrangement.

Hardware & rear chassis

Complete rear-access joints before the monitor makes them difficult to reach

Hardware	Qty	Where it goes
M3 x 12 socket-cap machine bolts	48	28 rear-splice + 4 bottom-shoe + 16 structural trim
M3 x 10 low-profile button-head bolts	2	Right-side service-cover retention
M3 hex nuts (5.5 mm AF, ~2.4 mm thick)	50	Machine-bolt joints, including captive trim/cover nuts
M3 washers	Up to 64	Rear splice/shoe joints; none in driver tunnels or carrier corners
M3 x 10 screws for plastic	10	2 rear posts + 4 side-stop + 4 cover-hook screws
M3 x 8 screws for plastic	15	6 exit inserts + 4 cover-splice + 1 anti-lift + 4 carrier
Wall-appropriate screws/anchors	6 sets	4 keyhole hangers + 2 anti-lift bracket wall fixings

For each raceway section add 6 M3 x 10 plastic screws for its cap and 2 suitable wall fixings. Plastic pilots are 2.7 mm: use the specified plastic-fastening screws; do not substitute longer screws into blind holes.

Tools

A slender 2.5 mm hex driver with at least 55 mm useful reach must fit the 6.5 mm rear trim tunnels. Match the key to your actual cover button-head screws (commonly 2 mm). Use hand pressure when tightening printed joints.

3. Join the rear chassis

Lay out the six tiles as shown on page 1. Align the seven **02-chassis-splice** plates over matching internal seam holes. Fit four M3 x 12 bolts per splice from the rear, with nuts inside. Start all joints loosely, bring the seams flush on a flat surface, then snug them evenly.

4. Add monitor supports

Fit both **03-monitor-bottom-shoe** parts with four M3 x 12 bolts total. Fit both **04-monitor-rear-post** parts with two M3 x 10 plastic screws total. Fit the left/right side stops with four M3 x 10 plastic screws total, entering from the rear.

Monitor, Pi & hidden wiring

Keep all connector loads off the display and circuit board

5. Seat the monitor and secure the trim

Load the 16 captive trim nuts through the interior-facing pocket openings. Put soft pads on the bottom shoes and rear supports. Seat the display on its housing, add only the rear shims needed, and retain pads/shims so they cannot fall out. Add soft top packing to limit upward housing travel to roughly 0.5-1 mm without clamping the glass.

Support the monitor while attaching **07-front-top-left/right**, **08-front-bottom-left/right**, and **09-front-left-lower/upper** with the 16 rear-access M3 x 12 bolts. These are structural trim fasteners; they do not hold the separate bay lid.

6. Install the Pi carrier

Install **06-open-pi-carrier** in the right bay using four front-accessible M3 x 8 plastic screws. Secure the existing black case through its plastic mounting ears/holes and the carrier slots. Keep straps off the PCB, GPIO pins and heatsink.

7. Connect before closing

Connect monitor mini-HDMI to the Pi HDMI adapter; connect monitor touchscreen USB-C to a Pi USB data port. Connect the monitor and Pi to their respective power supplies. Audio/Ethernet are optional if used. Keep the large adapters in the Pi carrier's open LEFT and BOTTOM corridors; route slack within the covered bay without pulling on ports.

The case allowance is 74 x 102 x 30 mm. Left adapter allowance is 45 mm; bottom allowance is 60 mm. These are design reserves, not measurements of every adapter. The left corridor extends behind the monitor: disconnect those plugs before pulling the Pi carrier forward.

Build & operate the service lid

Two right-side screws release the lid; the monitor stays installed

8. Assemble the lid off the enclosure

Join **10-right-bay-cover-1** (lower) and **10-right-bay-cover-2** (upper) using two **11-bay-cover-splice** plates on the hidden rear face and four M3 x 8 plastic screws. Fit two separate **11-bay-cover-hook** parts to the upper cover using four M3 x 10 plastic screws. Load the two M3 nuts into the side-retention blocks from their rear openings.

9. Close the bay

With wiring clear of the seating edges, hold the joined lid 8 mm below its final position. Push it back into place, then slide it **UP 8 mm** to engage both upper hooks. Fit the two M3 x 10 low-profile button-head bolts through the recessed RIGHT-side holes.

Routine Pi service

Power down and disconnect the power supplies. Remove both right-side cover screws. Ease the lid slightly forward (nominal 0.25 mm take-up), slide it DOWN 8 mm, then pull it forward. The hooks and backing plates stay attached to the lid.

For Pi removal, disconnect its cables first, remove the four front-accessible carrier screws, then withdraw the carrier/case forward. Do not drag connected left-facing adapters into the underside of the monitor. Refit the carrier and connections, then reinstall the lid in reverse order.

Keep the refrigerator clear of the RIGHT-side screw/driver path. Leave space below the lid for its 8 mm downward travel. The bay lid must move without snagging wires.

Wall mounting & final check

Cover the route to the outlet and preserve access for future service

10. Select the cable exit

Fit **13-exit-cable** at the chosen outlet route and **13-exit-blank** at the other two exits. Use six M3 x 8 plastic screws total. Fit **14-wall-cable-channel** sections and matching **15-wall-cable-channel-cap** sections along the route. Leave the power adapters accessible at the outlet; the enclosure does not combine their power supplies.

11. Mount beside the fridge

Attach the four **12-wall-spacer** parts to the chassis rear with thin adhesive strips clear of their slots. Position wall fixings from the actual assembled keyholes, using anchors appropriate to the wall and total load. Use the printed chassis as the alignment reference; the diagrams in this guide are not drilling templates.

Hang the supported enclosure on the four reinforced keyholes. Install **12-wall-anti-lift-bracket** with its two wall fixings and one underside M3 x 8 plastic screw into the chassis. Preserve 12 mm upward clearance for later lift-off and right-side driver access for the lid.

The modeled keyhole has a 10.5 mm head entry and 5.5 mm neck. Choose screw heads that retain across the neck and fit the entry (design head envelope up to 9.5 mm; shank up to 5 mm). Check actual head engagement before releasing the enclosure.

12. Before leaving it mounted

Confirm the bottom shoes support the housing; the glass is not under preload; the Pi case is secure; every plug is accessible with the bay open; wires clear the lid hooks and screws; ventilation slots remain open; and the wall mount cannot lift free with the anti-lift screw installed. Power up and check display, touch, and enclosed operation before considering installation complete.

Monitor replacement

Disconnect power, support the enclosure, remove the underside anti-lift screw, lift the enclosure about 12 mm and pull it off the wall. Reach the trim screws from the rear, supporting the monitor as retention is released.

Guide scope and recovered source

Based on the original REV8 README, its final mechanical-review record, retained CAD hardware notes, and part names read from the saved REV8 3MF. Physical fit, printed strength, wall capacity and operating temperature have not been certified. The saved Ready-to-Review project contains the cover hooks/splices; its object list does not include the optional 15-wall-cable-channel-cap. Obtain that matching cap if using the printed raceway. Current printer/print-completion status was not inspected for this guide.